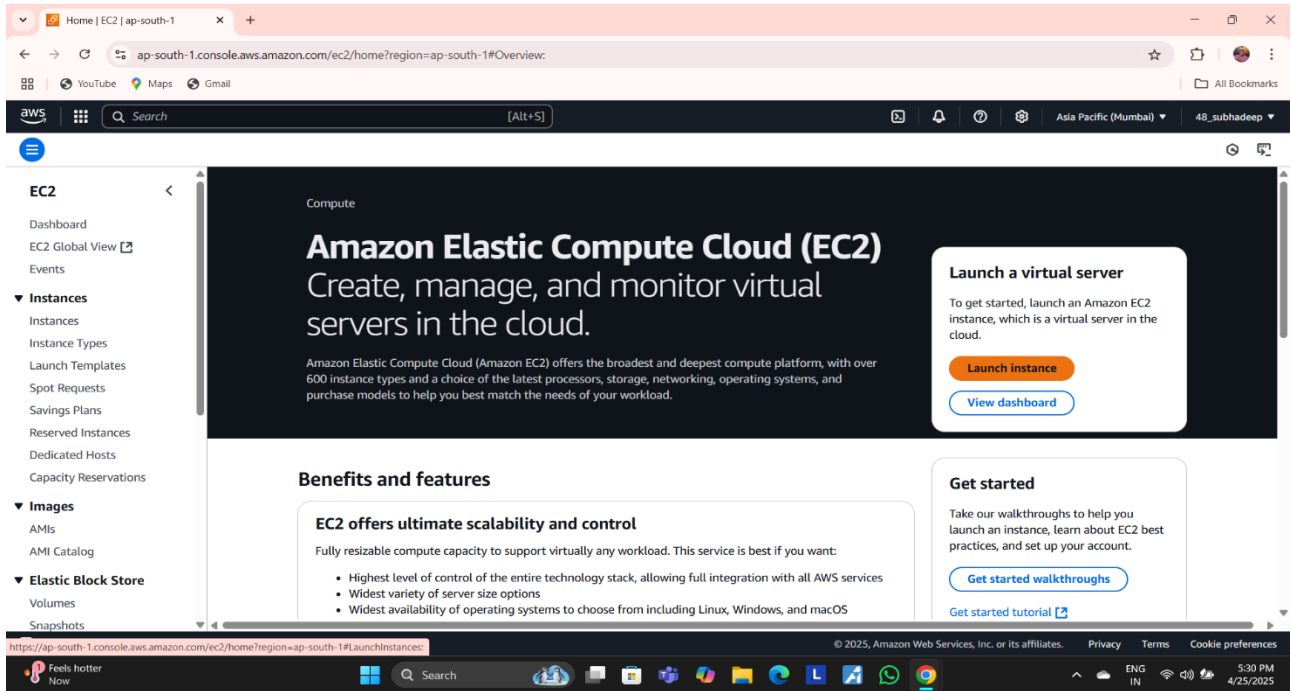


Assignment-12

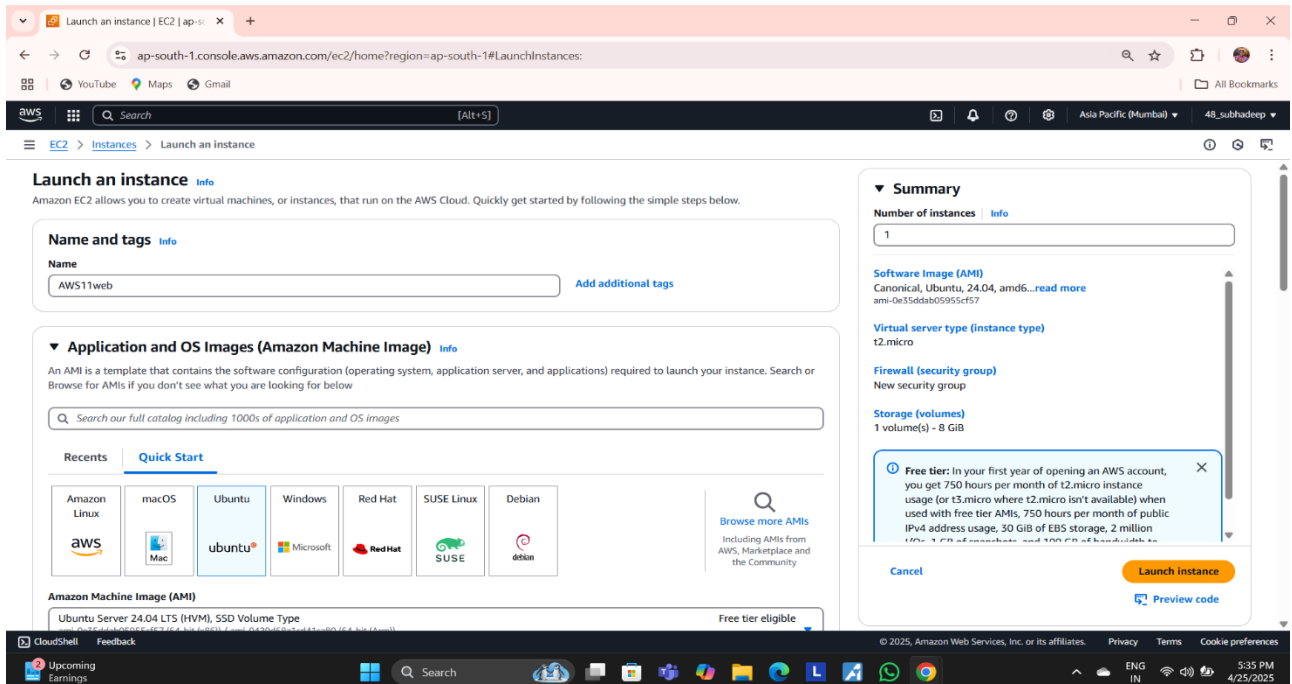
PROBLEM STATEMENT : Deploy and run the project in AWS without using the port.

Steps:

1. Go to the EC2 Dashboard.



2. Click on Launch instance.



Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Key pair name - required
spkey1 [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-055d721b1004494c1

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable
Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
[Create security group](#) [Select existing security group](#)

We'll create a new security group called 'launch-wizard-2' with the following rules:

- ☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0e35ddab05955cf57

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million IOPS, 1 GiB of ephemeral, and 100 GiB of bandwidth to S3.

[Cancel](#) [Launch instance](#) [Preview code](#)

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Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage [Info](#) [Advanced](#)

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier: eligible customers can get up to 30 GiB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

[Click refresh to view backup information](#)
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

Advanced details [Info](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
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[Cancel](#) [Launch instance](#) [Preview code](#)

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3. Instance is created successfully.

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Success
Successfully initiated launch of instance (i-02b3196dc4682dea2)

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute interval.
[Manage detailed monitoring](#)

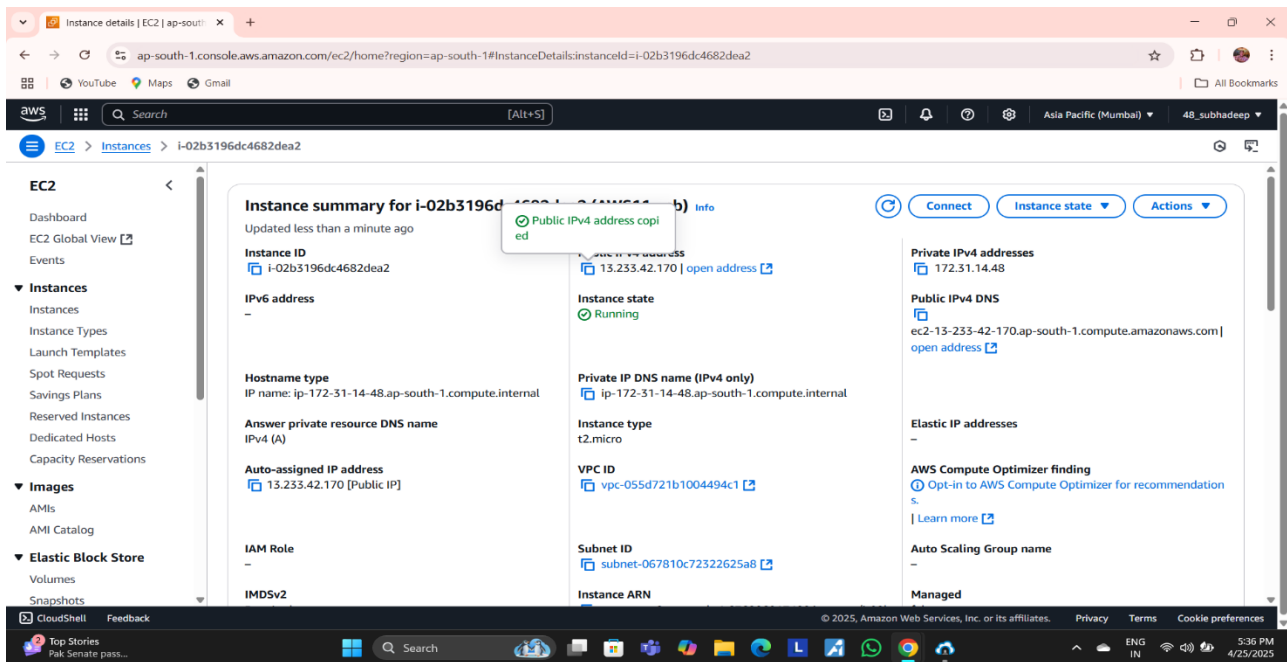
Create Load Balancer
Create an application, network gateway or classic Elastic Load Balancer.
[Create Load Balancer](#)

Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
[Create AWS budget](#)

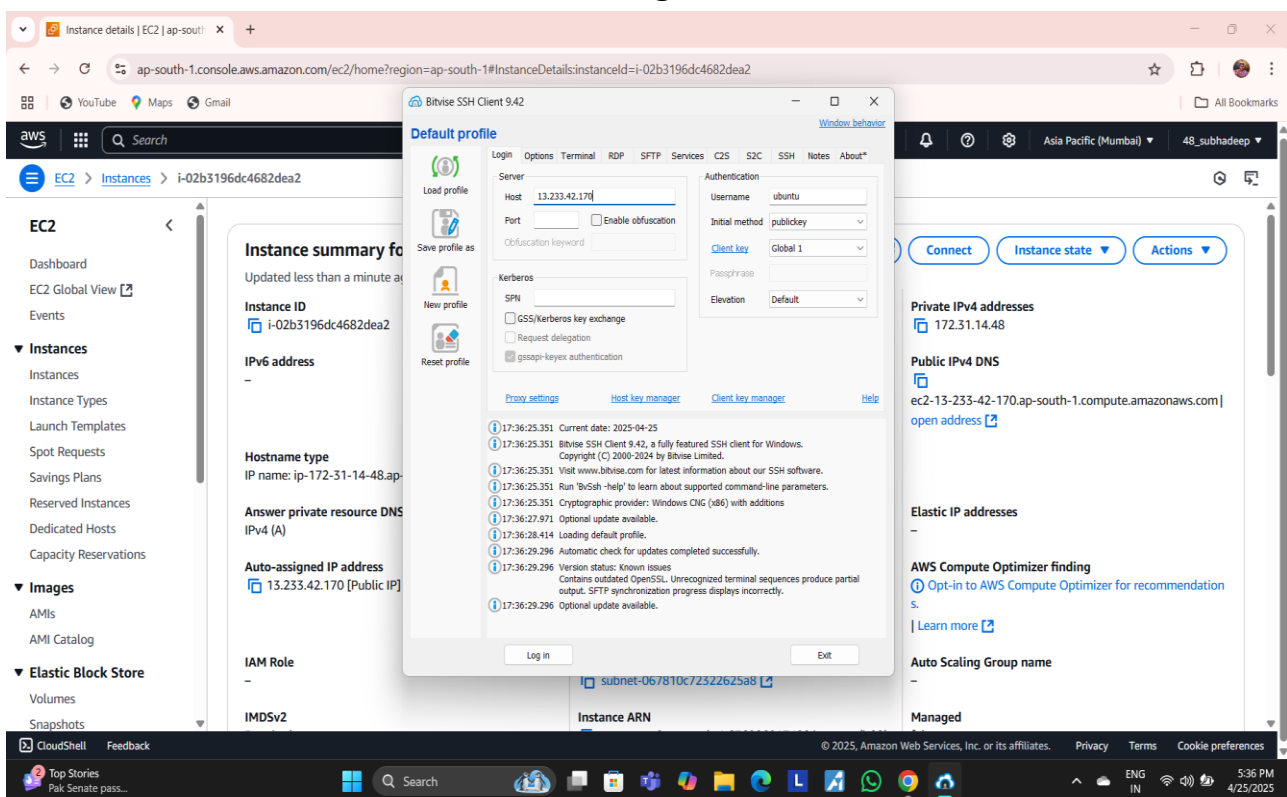
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
[Manage CloudWatch alarms](#)

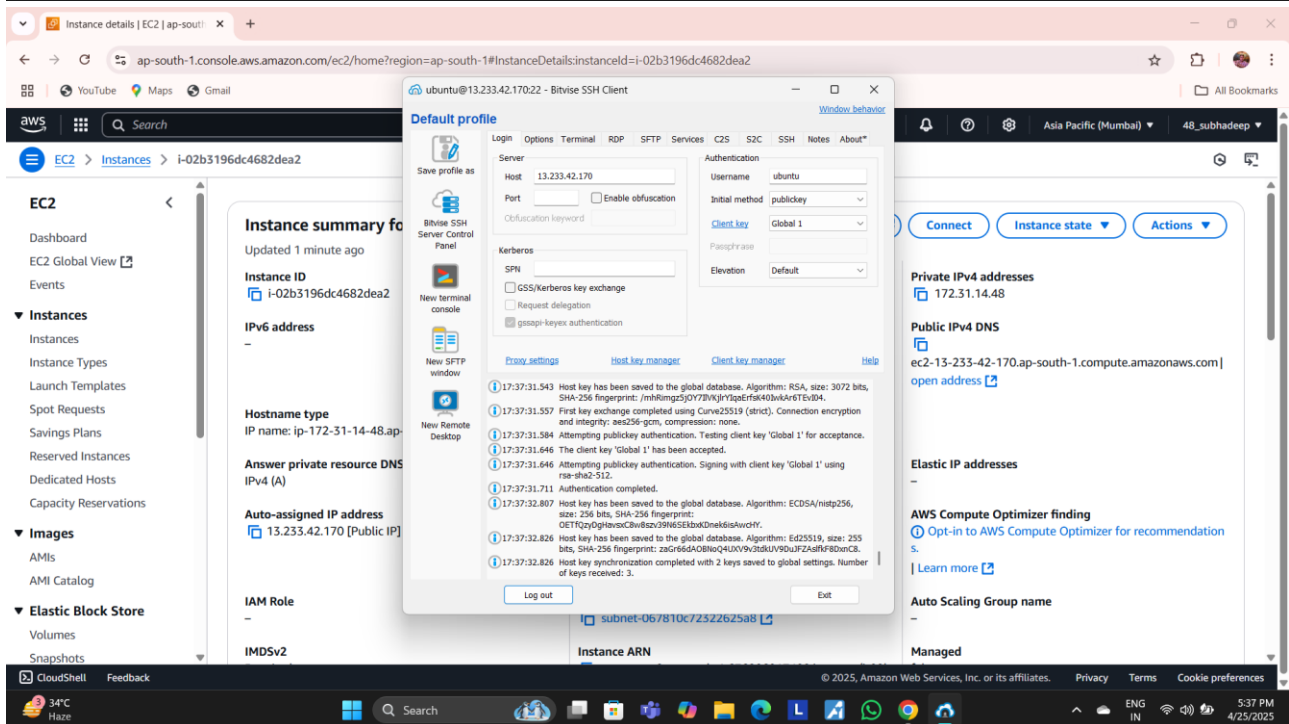
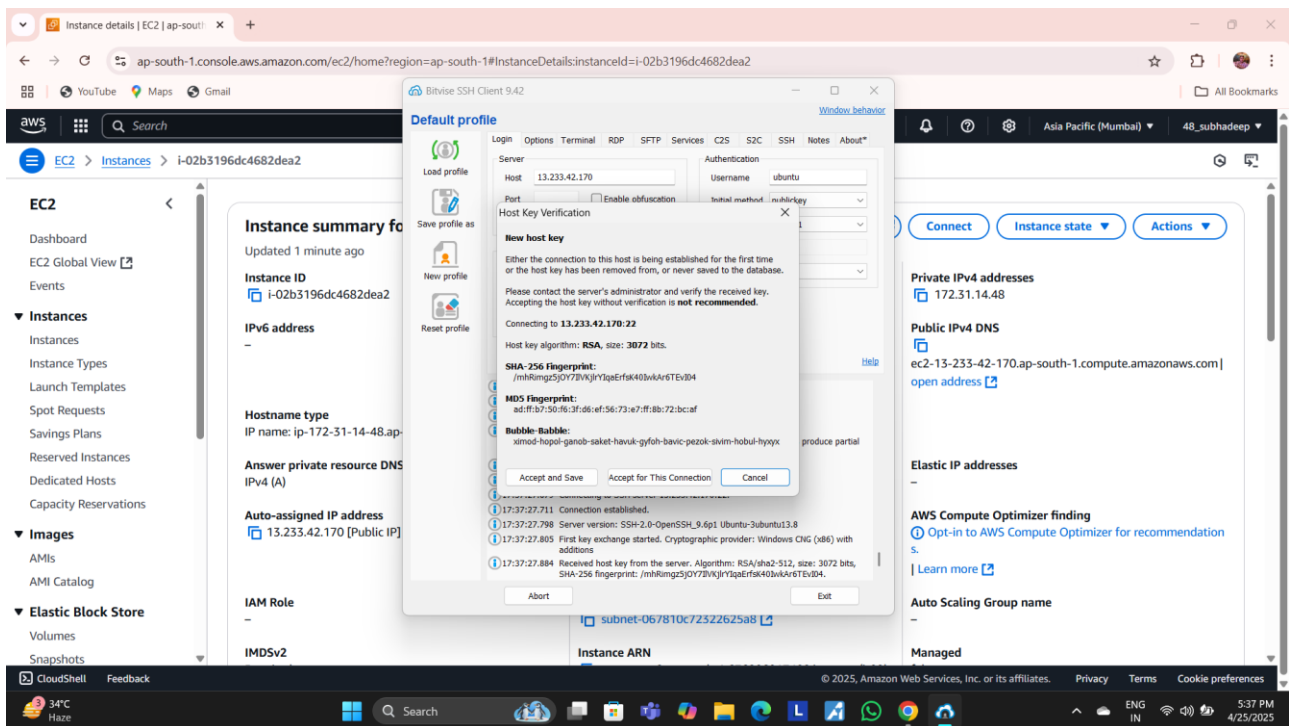
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4. Click on the instance ID and copy its public IPv4 address.



5. Paste into Bitvise SSH Client and Login.





6. Type the following commands in New SFTP Terminal:

```

sudo apt-get update
sudo apt-get upgrade
sudo apt-get install nginx
nginx -v curl -SL https://deb.nodesource.com/setup_18.x|sudo -E bash -
sudo apt install nodejs
git clone https://github.com/Subhadeep-pan/AWS_asg.git
npm install

```

7. Now in terminal. Type the following commands:

```
ubuntu@ip-172-31-14-48:~/AWS_asg$ cd/
-bash: cd/: No such file or directory
ubuntu@ip-172-31-14-48:~/AWS_asg$ cd /
ubuntu@ip-172-31-14-48:/$ ls
bin                dev                lib                lost+found         opt                run                snap               tmp
bin.usr-is-merged  etc                lib.usr-is-merged  media              proc              sbin              srv               usr
boot               home               lib64              mnt                root              sbin.usr-is-merged  sys               var
ubuntu@ip-172-31-14-48:/$ cd etc/
ubuntu@ip-172-31-14-48:/etc$ cd nginx/
ubuntu@ip-172-31-14-48:/etc/nginx$ ls
conf.d             koi-utf            modules-available  proxy_params       sites-enabled      win-utf
fastcgi.conf       koi-win            modules-enabled    scgi_params        snippets
fastcgi_params     mime.types         nginx.conf         sites-available    uwsgi_params
ubuntu@ip-172-31-14-48:/etc/nginx$ cd sites-available/
ubuntu@ip-172-31-14-48:/etc/nginx/sites-available$ ls
default
ubuntu@ip-172-31-14-48:/etc/nginx/sites-available$ sudo chmod 777 default
ubuntu@ip-172-31-14-48:/etc/nginx/sites-available$ sudo nano default
```

8. Now in the nano editor go to location and paste the following code:

```
proxy_pass https://localhost:4000;
proxy_https_version 1.1;
proxy_set_header Upgrade $http_upgrade;
proxy_set_header Connection 'Upgrade';
proxy_set_header Host $host;
proxy_cache_bypass $http_upgrade;
```

Now back to bash shell Type:

```
ubuntu@ip-172-31-14-48:/etc/nginx/sites-available$ cd ..
ubuntu@ip-172-31-14-48:/etc/nginx$ cd /
ubuntu@ip-172-31-14-48:/$ cd home
ubuntu@ip-172-31-14-48:/home$ ls
ubuntu
ubuntu@ip-172-31-14-48:/home$ cd ubuntu
ubuntu@ip-172-31-14-48:~$ ls
AWS_asg
ubuntu@ip-172-31-14-48:~$ cd AWS_asg
ubuntu@ip-172-31-14-48:~/AWS_asg$ ls
README.md  index.js  node_modules  package-lock.json  package.json
ubuntu@ip-172-31-14-48:~/AWS_asg$ node index.js
Started server
```

9. Now copy the public ipv4 address and paste in browser. It will land directly to html page.

Instance details | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-02b3196dc4682dea2

Search [Alt+S]

EC2 > Instances > i-02b3196dc4682dea2

Instance summary for i-02b3196dc4682dea2

Updated less than a minute ago

Instance ID
i-02b3196dc4682dea2

IPv6 address
-

Hostname type
IP name: ip-172-31-14-48.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
13.233.42.170 [Public IP]

IAM Role
-

IMDSv2

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-14-48.ap-south-1.compute.internal

Instance type
t2.micro

VPC ID
vpc-055d721b1004494c1

Subnet ID
subnet-067810c72322625a8

Instance ARN

Private IPv4 addresses
172.31.14.48

Public IPv4 DNS
ec2-13-233-42-170.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendation | Learn more

Auto Scaling Group name
-

Managed

Connect Instance state Actions

Public IPv4 address copied

13.233.42.170 | open address

CloudShell Feedback

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Hello MCKV, I am subhadeep